Divergent terms

1 Group I (with c.c.)

1.1 Row I

NA

1.2 Row II

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]\times$$
$$\left[\frac{11}{3}+\log\left(1-\frac{k^+}{\vee}\right)\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big] \tag{1.1}$$

1.3 Row III

NA

1.4 Row IV

NA

1.5 Row V

First part

NA

Second part

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\left[2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y). \tag{1.2}$$

The $\frac{11}{6}$ term of N_{2b} ("$N_{2b\,\mathrm{finite}}$" in the previous version) is removed from this file. It stays in the finite-terms file, where it is the $x\to z$ subtraction partner of the rest of N_{2b}; their sum contains no UV divergence and no $\log\mu$. Listing its UV value here as well would count it twice. The rest of N_{2b} would otherwise contain exactly minus this term.

2 Adding Group I

Rename $y\to x$, $z\to w$ in $N_{2b\,\mathrm{UV}}$. Since $\big[U(x')-U(w')\big]\big[U(x)-U(w)\big]=-\big[U(x')-U(w')\big]\big[U(w)-U(x)\big]$ and $\frac{g^4N_c}{16\pi^4}\big[2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}\big]=\frac{g^4N_c}{32\pi^4}\big[4\log(1-\frac{\Lambda}{k^+})-\frac{11}{3}\big]$:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma_E-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]$$
$$\times\left[\frac{11}{3}+\log\left(1-\frac{k^+}{\vee}\right)\right]\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big]$$
$$+\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma_E-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]$$
$$\times\left[4\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{3}\right]\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big] \tag{2.1}$$

Simplifying, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma_E-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]$$
$$\times\left[4\log\left(1-\frac{\Lambda}{k^+}\right)+\log\left(1-\frac{k^+}{\vee}\right)\right]\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right] \tag{2.2}$$

Setting $\Lambda\to 0$ and $\vee\to\infty$:

$$\frac{d^3N_{\rm NLO}}{d^3k}\Big|_{\text{Group I, divergent}} = 0.$$

So Group I has no net UV pole and no net $\beta_0\log\mu$: the $\frac{11}{3}$ of Row II cancels the $-\frac{11}{6}$ of $N_{2b\,\rm UV}$ (with c.c. on both). The running coupling comes entirely from Group II, Row IV (below).

3 Group II

3.1 Row I

3.1.1 First part

NA

3.1.2 Second part

NA

3.2 Row II

NA

3.3 Row III

NA

3.4 Row IV

$$\frac{d^3N_{\rm DGLAP}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\int_0^1\frac{d\xi}{\xi^2}\,\widetilde{\mathcal{P}}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4\,\xi^2}\right]\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{3.1}$$

where:

$$\widetilde{\mathcal{P}}_{gg}(\xi) = 2\,N_c\left[\xi(1-\xi)+\frac{\xi}{(1-\xi)_+}+\left[\frac{1-\xi}{\xi}\right]_+\right]+\frac{\beta_0}{2}\,\delta(1-\xi)$$

with, for $f(\xi)$ the full integrand to the right of $\widetilde{\mathcal{P}}_{gg}$ (including $1/\xi^2$),

$$\int_0^1 d\xi\,\frac{\xi}{(1-\xi)_+}f(\xi)\equiv\int_0^1 d\xi\,\frac{\xi f(\xi)-f(1)}{1-\xi},\qquad \int_0^1 d\xi\left[\frac{1-\xi}{\xi}\right]_+ f(\xi)\equiv\int_0^1 d\xi\,\frac{(1-\xi)f(\xi)-f(0^+)}{\xi},$$
$$f(0^+) = -\frac{4\pi}{k^2}\int_{x,x',z}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x).$$

$$\frac{d^3N_{\rm RC}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\beta_0}{16\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\left[2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x) \tag{3.2}$$

Running coupling. The $\frac{\beta_0}{2}\delta(1-\xi)$ part of DGLAP cancels RC exactly, so the row is unchanged by writing it this way. RC is the only $\beta_0\log\mu$ term in the whole result. Since $\frac{g^2\beta_0}{16\pi^2}=\frac{\alpha_s\beta_0}{4\pi}$ and the prefactor of RC is the LO one, LO + RC gives $\alpha_s(\mu)\big[1+\frac{\alpha_s\beta_0}{4\pi}\log\frac{(y'-y)^2\mu^2e^{2\gamma_E}}{4}\big]=\alpha_s\big(\frac{4e^{-2\gamma_E}}{(y'-y)^2}\big)$ inside the LO integral: one-loop running, $\beta_0=\frac{11N_c}{3}$ ($n_f=0$), at the parent dipole size. The $\log\mu$ in DGLAP is the fragmentation-scale dependence.

4 Group III (with c.c.)

4.1 Row I

NA

4.2 Row II

NA

4.3 Row III

4.3.1 First part

NA

4.3.2 Second part

NA

4.4 Row IV

NA

4.5 Row V

NA

4.6 Row VI

NA